



State of California
Regional Water Quality Control Board
**APPLICATION/REPORT OF WASTE DISCHARGE
GENERAL INFORMATION FORM FOR
WASTE DISCHARGE REQUIREMENTS OR NPDES PERMIT**

**I. FACILITY INFORMATION****A. Facility:**

Name: National Park Service - Clover Creek Wastewater Treatment Facility			
Address: 64140 Clover Creek Drive			
City: Sequoia National Park	County: Tulare	State: CA	Zip Code: 93271
Contact Person: Bright Avusuglo-Ahia		Telephone Number: (559) 565-3749	

B. Facility Owner:

Name: Sequoia and Kings Canyon National Parks			Owner Type (Check One) 1. ☐ Individual 2. ☐ Corporation 3. ☒ Governmental Agency 4. ☐ Partnership 5. ☐ Other: _____	
Address: 47050 Generals Highway				
City: Three Rivers	State: CA	Zip Code: 93271		
Contact Person: Bright Avusuglo-Ahia		Telephone Number: (559) 565-3749	Federal Tax ID:	

C. Facility Operator (The agency or business, not the person):

Name: Sequoia and Kings Canyon National Parks			Operator Type (Check One) 1. ☐ Individual 2. ☐ Corporation 3. ☒ Governmental Agency 4. ☐ Partnership 5. ☐ Other: _____	
Address: 47050 Generals Highway				
City: Three Rivers	State: CA	Zip Code: 93271		
Contact Person: Bright Avusuglo-Ahia		Telephone Number: (559) 565-3749		

D. Owner of the Land:

Name: Sequoia and Kings Canyon National Parks			Owner Type (Check One) 1. ☐ Individual 2. ☐ Corporation 3. ☒ Governmental Agency 4. ☐ Partnership 5. ☐ Other: _____	
Address: 47050 Generals Highway				
City: Three Rivers	State: CA	Zip Code: 93271		
Contact Person: Woody Smeck		Telephone Number: (559) 565-3101		

E. Address Where Legal Notice May Be Served:

Address: 47050 Generals Highway		
City: Three Rivers	State: CA	Zip Code: 93271
Contact Person: Woody Smeck		Telephone Number: (559) 565-3101

F. Billing Address:

Address: 47050 Generals Highway		
City: Three Rivers	State: CA	Zip Code: 93271
Contact Person: Justin Parone		Telephone Number: (559) 565-4460

CALIFORNIA ENVIRONMENTAL
PROTECTION AGENCY



State of California
Regional Water Quality Control Board

**APPLICATION/REPORT OF WASTE DISCHARGE
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II. TYPE OF DISCHARGE

Check Type of Discharge(s) Described in this Application (A or B):

☒ **A. WASTE DISCHARGE TO LAND**

☐ **B. WASTE DISCHARGE TO SURFACE WATER**

Check all that apply:

- | | | |
|--|--|---|
| ☒ Domestic/Municipal Wastewater Treatment and Disposal | ☐ Animal Waste Solids | ☐ Animal or Aquacultural Wastewater |
| ☐ Cooling Water | ☐ Land Treatment Unit | ☐ Biosolids/Residual |
| ☐ Mining | ☐ Dredge Material Disposal | ☐ Hazardous Waste (see instructions) |
| ☐ Waste Pile | ☐ Surface Impoundment | ☐ Landfill (see instructions) |
| ☐ Wastewater Reclamation | ☐ Industrial Process Wastewater | ☐ Storm Water |
| ☐ Other, please describe: _____ | | |

III. LOCATION OF THE FACILITY

Describe the physical location of the facility.

1. Assessor's Parcel Number(s)
Facility: N/A
Discharge Point: N/A

2. Latitude
Facility: 36° 36' 20.46"
Discharge Point: 36° 35' 53.11"

3. Longitude
Facility: 118° 45' 00.29"
Discharge Point: 118° 46' 22.17"

IV. REASON FOR FILING

- | | |
|---|--|
| ☐ New Discharge or Facility | ☐ Changes in Ownership/Operator (see instructions) |
| ☐ Change in Design or Operation | ☐ Waste Discharge Requirements Update or NPDES Permit Reissuance |
| ☐ Change in Quantity/Type of Discharge | ☒ Other: <u>Permitting Clover Creek WWTF under the General Order</u> |

V. CALIFORNIA ENVIRONMENTAL QUALITY ACT (CEQA)

Name of Lead Agency: United States Department of the Interior - National Park Service

Has a public agency determined that the proposed project is exempt from CEQA? ☒ Yes ☐ No

If Yes, state the basis for the exemption and the name of the agency supplying the exemption on the line below.

Basis for Exemption/Agency: Categorical exemption for existing facilities - Section 15301, USDI-NPS

Has a "Notice of Determination" been filed under CEQA? ☐ Yes ☒ No

If Yes, enclose a copy of the CEQA document, Environmental Impact Report, or Negative Declaration. If no, identify the expected type of CEQA document and expected date of completion.

Expected CEQA Documents:

☐ EIR ☐ Negative Declaration

Expected CEQA Completion Date: _____

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VI. OTHER REQUIRED INFORMATION

Please provide a COMPLETE characterization of your discharge. A complete characterization includes, but is not limited to, design and actual flows, a list of constituents and the discharge concentration of each constituent, a list of other appropriate waste discharge characteristics, a description and schematic drawing of all treatment processes, a description of any Best Management Practices (BMPs) used, and a description of disposal methods.

Also include a site map showing the location of the facility and, if you are submitting this application for an NPDES permit, identify the surface water to which you propose to discharge. Please try to limit your maps to a scale of 1:24,000 (7.5' USGS Quadrangle) or a street map, if more appropriate.

VII. OTHER

Attach additional sheets to explain any responses which need clarification. List attachments with titles and dates below:

You will be notified by a representative of the RWQCB within 30 days of receipt of your application. The notice will state if your application is complete or if there is additional information you must submit to complete your Application/Report of Waste Discharge, pursuant to Division 7, Section 13260 of the California Water Code.

VIII. CERTIFICATION

"I certify under penalty of law that this document, including all attachments and supplemental information, were prepared under my direction and supervision in accordance with a system designed to assure that qualified personnel properly gathered and evaluated the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment."

Print Name: Bright Avusuglo-Ahia

Title: Public Health Specialist

Signature: 

Date: December 2, 2019

FOR OFFICE USE ONLY

Date Form 200 Received:	Letter to Discharger:	Fee Amount Received:	Check #:
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UNITED STATES DEPARTMENT OF THE INTERIOR
NATIONAL PARK SERVICE

SEQUOIA AND KINGS CANYON NATIONAL PARKS

REPORT OF WASTE DISCHARGE

CLOVER CREEK WASTEWATER TREATMENT FACILITY
TULARE COUNTY, CALIFORNIA



December 2019

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INTRODUCTION

The United States Department of Interior, National Park Service, Sequoia and Kings Canyon National Parks currently operates the Clover Creek Wastewater Treatment Facility (WWTF) under Waste Discharge Requirements Order No. 88-031 (WDR 88-031) regulated by the Central Valley Regional Water Quality Control Board (Central Valley Water Board).

The purpose of this report is to renew WDR 88-031 pursuant to a request by the Central Valley Water Board to conform to its current plans and policies. The Report of Waste Discharge (ROWD) provides information of the WWTF for a new permit evaluation under the *State Water Resources Control Board Order WQ 2014-0153-DWQ - General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems (General Order)*.

1. BACKGROUND

1.1. Wastewater System Description

The Clover Creek WWTF is an indoor activated sludge and tertiary treatment facility initially constructed in 1986. The WWTF currently has a design capacity of 170,000 gallons per day (gpd) during the summer months and a winter flow of 82,000 gpd.

The WWTF is currently in design for a significant rehabilitation to become a plant with a peak capacity of 95,000 gpd. This is based on the past five years of wastewater demand, and will satisfy the General Order flow rate of 100,000 gpd. The original WWTF peak capacity was designed for full concessions build-out in 1996, but only a part of the original concession facilities were built with no additional facilities added to date.

The WWTF discharges disinfected tertiary-treated wastewater to land: sprayfields in the summer and subsurface leachfields in the winter. Wastewater is generated primarily from the over 900,000 park visitors annually. This includes wastewater from campground comfort stations, employee housing units, concessionaire facilities, a post office, a recreational vehicle (RV) station, and Wuksachi village within the Lodgepole area of the Sequoia and Kings Canyon National Parks.

Location and Facility Information

Attachment A of the ROWD shows the location and facility map.

The Clover Creek WWTF is located on the north side of the Generals Highway near Clover Creek and in Section 20, T15S, R30E, Mount Diablo Base and Meridian (MDB&M). The effluent disposal facilities are on the opposite site of the highway in the Red Fir area and in the NE 1/4 of Section 25, T15S, R29E, MDB&M and the NW of 1/4 of Section 30, T15S, R30E, MDB&M. The effluent treatment components consist of a septicage receiving station, preliminary treatment (manual screening and grinding, a

Parshall flume with ultrasonic flow sensor, equalization basins), secondary treatment (aeration basins and clarifiers), and tertiary treatment (disinfection facilities utilizing sodium hypochlorite). The WWTF has a wastewater solids handling unit that include aerobic digestion, indoor dewatering vacuum-assisted drying beds, and outdoor sand drying beds. Treated effluent is pumped to a manhole at the high point in the effluent line to break pressure and then flows by gravity to the storage tanks and leachfields/sprayfields.

There are no domestic groundwater wells located within 1/4 mile of the effluent disposal area. The disposal field is at an average elevation of 6,500 feet above mean sea level. The disposal area generally slopes at 5-18 percent in northwest and southwest directions, trending east to west. The vegetation of the disposal field varies from large trees (up to 48" diameter at breast height) to light ground vegetation. The surface water bodies about a 1/4-mile from the wastewater facilities are the Clover Creek and Halstead Creek. These creeks are tributaries to the Marble Fork and that of the Kaweah River.

Potable Water Supply

The National Park Service (NPS), Wolverton community water system (CWS) serves employee housing, commercial developments, campgrounds, WWTF, and maintenance facilities. The CWS consists of Wolverton and Lodgepole water treatment plants that allow for the distribution systems to operate together or separately for operational and seasonal flexibility.

- **Wolverton Water System**

The Wolverton water system obtains surface water from the Wolverton Creek and Well No. 1. The system consists of a 20,000-gallon sedimentation tank, a slow sand filter, a treatment plant with MIOX® chlorinator, a 100,000-gallon underground concrete tank, a pipeline, two 175,000-gallon above ground welded steel storage tanks located at Wuksachi with an additional chlorine dosing station and two-way solenoid valve, and a distribution system.

Raw surface water from the Wolverton Creek flows through slow sand filtration system and then flows by gravity to the treatment plant. Water from the Well No.1 is pumped by a submersible pump with a motor through a separate water main with a pitless adapter to the treatment plant. The surface water and ground water well piping are connected in series in the water treatment plant and controlled by programmable logic control (PLC) via set points from a level transducer in the storage tank.

In the treatment plant, the surface water flows through an inline water filter, a water softener, a 55-gallon brine tank, a MIOX® on-site generation system powered by electricity, and a mixed oxidant solution tank to provide disinfection. Groundwater from Well No.1 is combined with surface water before disinfection. The free chlorine residual

is continuously monitored by a Hach® CL17chlorine analyzer. From the plant, water is pumped to the 100,000-gallon storage tank where it achieves minimum contact time.

The finished water from the 100,000-gallon tank flows through an 8-inch and a 4-inch water main by gravity and serves Wolverton, Upper Sherman Tree, and Wolverton Corrals. The Giant Forest water system has a groundwater well with secondary source water available from the Wolverton water treatment plant in case of an emergency such as well failure.

A separate 6-inch water main is connected to the 8-inch water main to provide water to Lodgepole, Wuksachi, and Red Fir. From the 6-inch pipeline, a 3-inch water main is directly interconnected through a Cla-Val pressure reducing valve to provide an emergency water supply to the Lodgepole water distribution system.

From the 6-inch water main, water flows to a chlorine dosing station. In the dosing station, an LMI metering pump is utilized to inject National Sanitation Foundation (NSF) Standard 60 certified liquid sodium hypochlorite from a 35-gallon storage tank into the water before it flows into the two-175,000-gallon tanks and distributes water by gravity to Wuksachi and Red Fir communities. Only one 175,000-gallon tank is in operation, meeting peak demand for potable and fire storage.

- Lodgepole Water System

The Lodgepole water system obtains surface water from Sillman Creek. The water system has a diversion dam and intake structure, sand settling box, a 200,000-gallon underground concrete tank (clearwell), a treatment plant, two inline multi-media filters, and a distribution system.

Raw surface water from the screened intake structure flows through a 4-inch water main and sand settling box by gravity to the treatment plant. In the treatment plant, the treatment process is initiated by injecting NSF certified polymer coagulant (Sanafloc 5569p) from a 30-gallon LMI tank with metering pump into the raw water, upstream of the multi-media pressure tanks that operate in parallel. The polymer is added before water enters a static inline mixer. Filtration treatment is achieved by the two multi-media pressure filters. The filter backwash cycle is initiated manually by the operators.

In the plant, water is injected with food grade, NSF-approved Trisodium Phosphate, Crystalline (TSPC) to provide corrosion control treatment. Also in the plant, an LMI® metering pump is utilized to inject NSF Standard 60 certified liquid sodium hypochlorite from a 35-gallon storage tank into the water. The free chlorine residual is continuously monitored a Hach® CL17chlorine analyzer. From the treatment plant, water flows to the 200,000-gallon storage tank where it achieves minimum contact time. The water level is controlled by mechanical float valve in a vault near the storage tank. The distribution system is gravity flow, with water pressure between 45-60 pounds per square inch.

1.2. Service Area Description

Existing Regional Collection System

None

Wastewater Collection System

The sewer collection system consists of two areas: Lodgepole and Clover Creek. The sewer collection system in the Lodgepole area was constructed in 1940. In 1983, the system was modified to allow sewage flow to the lift stations before pumping to the Clover Creek WWTF. The sewers in the Clover Creek area were constructed in 1986 and a number of improvements were made at that time to the Lodgepole area collection system to reduce infiltration and inflow. The collection system is wastewater only and not combined with stormwater. The collection system serves the Lodgepole and Clover Creek areas and consists of 4-inch, 6-inch, 8-inch, and 10-inch ductile iron, cast iron cured-in-place pipe liners, and polyvinyl chloride gravity sewers and force mains of various lengths.

Attachment B of the ROWD shows the collection system layout.

The collection system includes two (2) lift stations to support both summer and winter operations. These lift stations are Lodgepole (primary) and RV Housing Area, located near the Clover Creek WWTF. The gravity sewer system collects wastewater flow from the Lodgepole area including employee housing, post office, concessionaire facilities, campground comfort stations, service station, and maintenance facilities of the park. The primary lift station receives wastewater from the Lodgepole area and pumps it through the force main to the WWTF headworks. The gravity sewer system collects wastewater flows from the Clover Creek area including Wuksachi Village and the RV housing area lift station pumps sewage from the parking areas through the force main into the gravity system where it flows to the WWTF headworks.

The collection system is well maintained and was hydro-flushed in 2010, and some other sections of the system are planned for 2025. Most existing manholes and sewer lines in the collection system were replaced between 2011 and 2013. In 2013, comprehensive manhole inspection and sealing was performed on the sewer system to minimize rain and snowmelt inflow and infiltration into the sewer system. Replacement of pumps, motors, and controls for the Lodgepole lift station was completed in 2015. The NPS wastewater operators oversee the maintenance of the collection system for overflows at the manholes and the influent lift stations. The NPS staff provides immediate response in the event of a sewer spill.

Storm Water Collection System

None

2. WASTEWATER CHARACTERIZATION AND TREATMENT

2.1. Domestic Wastewater Characterization (Untreated Wastewater)

Generation of Wastewater

In general, wastewater characteristics are typical of standard residential sources. The collection system has septage receiving station near the WWTF and RV stations. There are no industrial sources. There is only one commercial kitchen which has a grease interceptor.

Wastewater Flow Rate

Wastewater influent flow is measured by an ultrasonic flow sensor located at the headworks on the pipe near a Parshall flume. Due to surface flooding, seasonal snowmelt and thunderstorms have the potential to cause high wastewater flow into the Clover Creek wastewater system.

The monthly average flows at the WWTF varied from 10,800 to 59,185 gpd from January 2016 through December 2018. The influent flows at the WWTF are tabulated in Table 1.

Table 1. Wastewater Influent Flows

Month	Units	2016			2017			2018		
		Min	Ave	Max	Min	Ave	Max	Min	Ave	Max
January	gpd	10,000	23,267	36,000	19,411	98,396	997,616	10,042	17,067	28,646
February	gpd	14,000	27,036	51,000	0	54,721	387,011	10,621	20,508	33,460
March	gpd	15,000	36,567	69,000	21,228	42,570	71,629	16,705	45,654	259,260
April	gpd	12,000	34,414	71,000	36,263	59,185	119,773	20,404	35,895	68,890
May	gpd	0	34,567	69,000	24,478	42,889	114,095	20,662	30,277	49,270
June	gpd	20,000	37,300	55,000	22,201	41,402	60,808	28,055	39,139	54,720
July	gpd	25,000	42,767	60,000	20,513	111,285	826,491	27,779	46,806	54,978
August	gpd	19,000	36,612	59,000	21,821	41,321	59,514	20,406	45,190	77,781
September	gpd	0	29,134	69,800	13,283	34,709	52,261	9,100	17,900	43,200
October	gpd	11,134	20,637	34,412	15,540	22,656	29,679	8,900	14,800	28,000
November	gpd	11,957	21,176	29,036	7,378	17,330	32,687	6,300	10,800	17,200
December	gpd	9,256	25,215	85,288	6,648	15,284	24,570	10,476	21,897	37,954
Limit: 30-Day	gpd	170,000	170,000	170,000	170,000	170,000	170,000	170,000	170,000	170,000

Notes:

- 1 Maximum influent flow values for January, February, and July 2017 were high as a result of a thunderstorm event.
- 2 Maximum influent flow value for March 2018 was high as a result of a thunderstorm event.

Clover Creek Wastewater Treatment Facility
Report of Waste Discharge

Wastewater Characteristics

From June to July 2019, wastewater influent and effluent samples collected at the WWTF to characterize nitrogen are tabulated in Table 2.

Table 2. Influent and Effluent Nitrogen Data

	Constituent	Units	6/5/2019	6/26/2019	7/2/2019	7/31/2019
Influent	Nitrate + Nitrite as N	mg/L	1.6	0	2.5	2.4
	Total Kjeldahl Nitrogen	mg/L	48.9	74	73	74
	Ammonia as Nitrogen	mg/L	31	71	77	68
	Total Nitrogen ¹	mg/L	50.5	74.0	75.5	76.4
Effluent	Nitrate + Nitrite as N	mg/L	33.9	44.8	64.1	74.4
	Total Kjeldahl Nitrogen	mg/L	0	0	0	0
	Ammonia as Nitrogen	mg/L	0	4.8	0.8	0
	Total Nitrogen ¹	mg/L	33.9	44.8	64.1	74.4
	Nitrogen Reduction ²	%	33.0	40.0	16.0	3.0

Notes:

1 Total Nitrogen is calculated as the sum of Nitrate as N and Total Kjeldahl Nitrogen

2 Total Reduction is calculated $[(\text{Influent Total Nitrogen} - \text{Effluent Total Nitrogen}) / (\text{Influent Total Nitrogen})] \times 100$

From September to October 2019, recreational vehicle data as collected at the influent of the Lodgepole RV Dump station are tabulated in Table 3.

Table 3. Recreational Vehicle Discharge Data

Constituent	Units	9/4/2019	9/18/2019	9/25/2019	10/16/2019	10/23/2019
Zinc	mg/L	0.71	2.44	0.43	0.39	0.42
Phenol	mg/L	ND	ND	2.40	2.50	ND
Formaldehyde	mg/L					

Notes:

1. ND - Not Detected

2. Formaldehyde samples collected by NPS were cancelled because the subcontract laboratory was not able to provide accurate data due to sample matrix. NPS will resample in Summer 2020.

Clover Creek Wastewater Treatment Facility
Report of Waste Discharge

In August 2019, electrical conductivity data as collected at the source water, influent, and effluent of the WWTF are tabulated in Table 4.

Table 4. Daily Electrical Conductivity Data

August 2019					
Date	Wolverton Well µmhos/cm	Wolverton Creek µmhos/cm	LP Sillman Creek µmhos/cm	WWTF Influent µmhos/cm	WWTF Effluent µmhos/cm
8/1/2019	76.2	70.2	12.6	1047.0	1271.0
8/2/2019	75.1	31.1	13.8	1035.0	1572.0
8/3/2019	80.0	37.3	12.2	1117.0	1243.0
8/4/2019	85.0	28.7	10.2	799.0	1701.0
8/5/2019	79.6	28.2	12.5	830.0	1999.0
8/6/2019	80.4	30.4	14.4	1017.0	1829.0
8/7/2019	79.0	31.6	11.6	1112.0	1645.0
8/8/2019	86.0	33.2	12.8	1107.0	2020.0
8/9/2019	80.8	28.7	16.4	1089.0	2480.0
8/10/2019	79.0	26.2	48.7	976.0	2360.0
8/11/2019	82.5	30.5	16.7	1118.0	2380.0
8/12/2019	79.1	27.7	10.5	958.0	2170.0
8/13/2019	78.4	25.6	11.3	946.0	2410.0
8/14/2019	80.9	38.4	14.1	880.0	1702.0
8/15/2019	82.3	27.3	15.9	1095.0	2100.0
8/16/2019	84.8	27.1	10.0	978.0	1983.0
8/17/2019	89.9	34.7	12.9	1110.0	2086.0
8/18/2019	76.9	32.7	12.7	918.0	1504.0
8/19/2019	80.9	27.6	20.6	2030.0	1698.0
8/20/2019	83.9	26.8	16.8	1263.0	1526.0
8/21/2019	79.0	28.0	18.5	1488.0	2216.0
8/22/2019	84.0	26.9	13.6	1229.0	1927.0
8/23/2019	83.0	28.0	14.2	836.0	2040.0
8/24/2019	83.2	28.2	12.5	938.0	2010.0
8/25/2019	79.9	33.0	12.4	1126.0	2116.0
8/26/2019	81.1	32.1	13.3	982.0	2001.0
8/27/2019	77.9	27.6	12.0	823.0	1676.0
8/28/2019	54.7	27.0	11.4	855.0	1057.0
8/29/2019	80.3	27.3	13.9	947.0	1272.0
8/30/2019	83.3	29.2	11.0	1042.0	1043.0
8/31/2019	77.8	27.9	12.4	987.0	1127.0
Minimum:	54.7	25.6	10.0	799.0	1043.0
Average:	80.2	30.9	14.6	1054.1	1811.7
Maximum:	89.9	70.2	48.7	2030.0	2480.0

2.2. Wastewater Treatment System

Wastewater treatment schematic

Attachment C of the ROWD shows the wastewater treatment flow schematic.

Wastewater Components and Treatment Technology

The WWTF had rehabilitation of mechanical and controls systems completed in 2015. A second improvement to the WWTF is currently under engineering design with anticipated construction in fall of 2021.

The WWTF improvements will convert the current coarse bubble with mixer aeration system to fine bubble aeration and will rehabilitate sludge handling. The following describes the plant as it is presently. As shown in the attached flow schematic, the indoor treatment process at the Clover Creek WWTF is as follows:

- Septage Receiving Station – Consists of a bar screen, blower, and grinder pumps.
- Headworks – Consists of an inclined bar screen, drain tray, grinder, and a Parshall flume with ultrasonic level sensor.
- Equalization – Two aerated equalization tanks with soda ash dosing.
- Aeration Basins – Four aeration basins with coarse bubble mixer aeration.
- Secondary Clarifiers – Four rectangular clarifiers.
- Disinfection System – Hypochlorite disinfection with chlorine contact basins.
- Effluent Storage – Two effluent storage tanks continuously controlled by transducer.
- Effluent Disposal – Treated effluent flows by gravity to sprayfields and leachfields.
- Digester Basins – Five aerobic digesters with coarse bubble mixer aeration.
- Sludge Dewatering – Two indoor vacuum-assisted drying beds and four outdoor sand drying beds to which sludge is hauled off-site.

Septage Receiving Station

The septage receiving facility consists of a concrete storage tank with a manual bar screen and grinder pumps. The tank is equipped with coarse bubble mixer aeration to aerate the tank. Septage is pumped from the tank to the headworks and aerobic digesters utilizing two grinder pumps that are located in a small room attached to the storage tank. The total storage capacity of the tank is 5,000 gallons.

Headworks

The pre-treatment equipment is located in headworks building separated from the WWTF. The headworks consist of an inclined manual bar screen installed in the bypass channel and in-channel, drain tray, and grinder. Wastewater influent flows are measured downstream of the grinder with a Parshall flume and ultrasonic level sensor. A small positive displacement blower is also located in the headworks building. This blower also aerates the septage receiving tank to reduce odors.

Equalization

Two equalization tanks are utilized to balance influent flows. Each equalization tank is equipped with coarse bubble mixer aeration to aerate the tanks. Two positive-displacement blowers located in the blower room are assigned to the equalization tanks. Two air-diaphragm pumps are used to pump wastewater out of the equalization tanks to the aeration basins. The plant operators manually select a pumping rate to pump out of the equalization basins. The pumps automatically turn on and off based on the water level in the equalization tanks measured by a bubbler and pressure transducer. Influent knife gates located within the tanks are used to isolate and interconnect the tanks.

A soda-ash chemical feed system is used to add alkalinity at the equalization tanks to maintain an acceptable pH. Soda ash is manually batched in a soda ash feed tank. The tank is equipped with a mixer and a chemical feed pump to dose soda ash to the equalization tanks. The total storage volume of the equalization basin is 48,000 gallons.

Aeration Basins

Four aeration basins are available to treat influent flows. Presently, two aeration basins are used for treatment. The two idled aeration basins have not been used for the past 10 years and currently function as equalization basin overflow tanks.

Each aeration basin is equipped with coarse bubble mixer aeration to aerate the tanks. Four positive-displacement blowers located in the blower room are dedicated to the aeration basins, as well as a secondary dedicated backup blower. Mixed liquid flows by gravity from the aeration basins to the clarifiers.

A polymer chemical feed system is used to add polymer to the aeration basin influent. The polymer is manually batched in a polymer storage tank. A chemical metering pump is used to dose the polymer into the aeration basin influent. The total storage capacity of the aeration basin is 170,000 gallons.

Secondary Clarifiers

Four rectangular clarifiers are provided to settle mixed liquor from the aeration basins. Only clarifiers 1 and 2 are currently operational. Wastewater flows from the clarifiers by gravity to the chlorine contact basins. Each clarifier is equipped with a chain and flight type clarifier mechanism. Five recessed-impeller centrifugal pumps are provided for returning activated sludge back to the aeration basins and waste sludge to the digesters. A motor-operated two-way valve is provided on the common discharge of the pumps to direct the sludge to the aeration basins or the digesters.

Manually operated dipping weirs are provided to draw accumulated scum off of the wastewater side of the clarifiers. The scum flows by gravity to a scum pit. Two recessed-impeller centrifugal pumps are provided to pump accumulated scum to the digesters. The clarifiers have 72,000 gallons of storage capacity.

Disinfection System

Wastewater from the clarifiers flows to chlorine contact basins. Hypochlorite is dosed to provide disinfection. The original plant design included a chlorine gas feed system. For safety reasons, the NPS replaced the chlorine gas with an onsite hypochlorite generator in 2005. The hypochlorite generation system consists of a water softener, brine saturation tank, electrolytic cells, and three hypochlorite solution storage tanks. In addition to the hypochlorite generation equipment, one chemical metering pump is used to dose hypochlorite solution to the chlorine contact tank. The total chlorine contact tank is 10,900 gallons.

Storage Facilities

There are two steel effluent storage tanks located at Red Fir between the treatment plant and the disposal fields. Tank No. 1 is a 10,000-gallon storage tank and Tank No. 2 is a 250,000-gallon storage tank. Tank No. 2 is normally used for operations. Tank No. 1 is only used when the Tank No. 2 is taken out of service for cleaning or other maintenance. These storage tanks serve as surge storage when plant output intermittently exceeds disposal field dosing rates.

A magnetic flow meter is installed to measure treated effluent from the WWTF before it flows to the storage tanks and disposal facilities.

Neither the sprayfield nor the leachfield operates continuously. Each system operates intermittently at full design flow rate independently of the plant flow. The tank is connected with a side outlet tee of the effluent pipeline that extends to the leachfields and sprayfields. This connection allows partial treated effluent flow to the storage tank. When the disposal fields are operational, all of the treated effluent from the WWTF flows directly through the effluent pipeline to the disposal systems. When the disposal facilities are not operational, the entire treated effluent from the WWTF fills effluent pipeline between the manhole at high point and disposal system through diverted side outlet tee into the storage tank. The storage tank is controlled by pressure transducer located in a vault used to sense the level in the storage tank and transmit signal back to the WWTF.

Digester Basins

Five digester basins of varying volumes are available at the WWTP to store and treat waste activated sludge. Each digester basin is equipped with coarse bubble mixer aeration. Each digester is equipped with a telescoping valve to allow the digesters to be settled and the supernatant decanted off the top.

Four blowers are provided to supply air to the digesters. The discharge piping for the digester blowers is interconnected, which provides a lot of flexibility for air supply to the digesters. The discharge piping for the digester blowers is also interconnected with the discharge piping for the equalization tank blowers.

A single submersible pump is used to pump digested sludge from the digesters to the sludge drying beds. Two submersible pumps are provided to transfer decanted water from the digesters to the equalization tanks. These pumps also serve to pump collected water from the sludge dewatering process back to the equalization tanks.

Aerobic digester basins 4 and 5 are functioning and are currently used. Digester basins 1, 2, and 3 are not in service and have not been used for a number of years. The total storage capacity of the digester is 81,000 gallons.

Sludge Dewatering

The WWTF has two sludge dewatering processes: two indoor vacuum-assisted dewatering beds and four outdoor sand drying beds.

The vacuum-assisted dewatering beds are not used. The beds and equipment are in good condition. If the vacuum beds need to be utilized, rehabilitation of the equipment would be anticipated.

The outdoor sand drying beds are currently used in dewatering the digested sludge. Sludge is pumped from the digesters into drying beds and allowed to drain by gravity and dry for approximately two weeks. Dried solids are manually raked and shoveled from the drying beds into containers stored outside. These containers are then picked up by a contract hauler and the sludge is hauled outside the park for composting.

The surface area of the indoor vacuum-assisted dewatering beds is 416 square feet and the surface area of the outdoor sand drying beds is 4,200 square feet.

Clover Creek Wastewater Treatment Facility
Report of Waste Discharge

Wastewater Effluent Quality

The monthly average effluent analysis of biochemical oxygen demand (BOD), total suspended solids (TSS), total coliform organisms, and settleable matter, based on the monthly Self-Monitoring Report (SMR) from January 2016 to December 2018, are tabulated in Table 5.

Table 5. Monthly Average Effluent Quality

	2016				2017				2018			
Month	BOD ₅ mg/L	TSS mg/L	Total Coliform MPN/100 mL	Settleable Matter mg/L	BOD ₅ mg/L	TSS mg/L	Total Coliform MPN/100 mL	Settleable Matter mg/L	BOD ₅ mg/L	TSS mg/L	Total Coliform MPN/100 mL	Settleable Matter mg/L
January	3.18	23.75	2.53	0	9.25	28.75	76.63	<0.1	2.04	5.80	<1.8	<0.1
February	2.35	35.00	<1.8	0	6.70	36.50	14.50	<0.1	2.03	8.25	<1.8	<0.1
March	1.74	16.40	2.34	0	4.83	25.00	6.90	<0.1	1.93	11.00	<1.8	<0.1
April	2.23	11.75	<1.8	0	14.15	39.50	85.90	0.19	1.65	11.75	5.75	<0.1
May	3.35	17.00	<1.8	0	16.40	30.80	191.80	0.25	2.42	11.60	<1.8	<0.1
June	11.48	41.40	349.26	0	19.80	64.50	86.00	0.11	4.65	10.75	<1.8	<0.1
July	6.98	18.75	476.50	0	10.40	42.25	287.50	2.21	5.30	24.75	103.45	<0.1
August	18.02	47.60	695.20	0	8.96	49.60	350.40	<0.1	10.46	17.20	19.08	<0.1
September	6.93	28.75	414.95	<0.1	6.28	31.50	14.15	0.13	3.50	16.75	21.15	<0.1
October	10.50	57.50	48.88	<0.1	4.93	13.50	<1.8	<0.1	1.60	8.80	391.08	<0.1
November	4.86	26.60	127.30	<0.1	4.94	10.20	2.80	<0.1	2.85	7.25	<1.8	<0.1
December	3.68	26.75	5.75	<0.1	2.10	8.50	<1.8	<0.1	2.33	9.75	2.48	<0.1
Limit: 30-Day	30	30	23	0.10	30	30	23	0.10	30	30	23	0.10
Limit: Inst Max	60	60	240	0.50	60	60	240	0.50	60	60	240	0.50

The TSS concentrations exceeded the monthly average limits of 30 mg/L in February, June, August, and October 2016, and February, April through September 2017. In June 2017, the TSS exceeded the instantaneous maximum limits of 60 mg/L, as described in WDR Order No. 88-031.

Total Coliform results for June through November 2016, January, April through August 2017, and July and October 2018 exceeded both monthly average limits of 23 Most Probable Number (MPN/100mL) and instantaneous maximum limits of 240 MPN/100mL. According to the SMR for July 2018, a power surge caused damage to blowers and operational compressors that affected the chlorine generation system.

Clover Creek Wastewater Treatment Facility
Report of Waste Discharge

Treated Effluent Disposal

The monthly average effluent flows at the WWTF ranged from 14,000 to 58,343 gpd from January 2016 through December 2018. The monthly effluent flows at the WWTF are tabulated in Table 6.

Table 6. Wastewater Average Effluent Flows

Month	Units	2016		2017		2018	
		Effluent Sprayfield (Summer) Ave	Effluent Leachfield (Winter) Ave	Effluent Sprayfield (Summer) Ave	Effluent Leachfield (Winter) Ave	Effluent Sprayfield (Summer) Ave	Effluent Leachfield (Winter) Ave
January	gpd		14,000		43,522		20,442
February	gpd		28,571		54,152		17,179
March	gpd		23,548		44,110		31,323
April	gpd		32,125		58,343		34,267
May	gpd		50,000		40,352		33,697
June	gpd	85,000	11,667	24,630	12,164	1,690	34,533
July	gpd	45,474		46,910		49,790	
August	gpd	37,739		53,903		22,101	
September	gpd	32,067		36,657		16,478	
October	gpd	6,795	8,167	24,837		23,895	
November	gpd		14,833	14,419	21,200	19,933	
December	gpd		16,639		18,158		20,355
Limit: 30-Day	gpd	170,000	82,000	170,000	82,000	170,000	82,000

Existing Sprayfield and Leachfield

The disposal facilities in the Red Fir area use sprayfields in the summer months and leachfields in the winter months. The disposal area has a propane storage tank, a building to accommodate a backup power generation system, and Tesco SCADA equipment.

The disposal facilities have a significant area to accommodate effluent disposal. The sprayfield occupies an area of over 23 acres and has 18 separate sprayfield zones. The disposal area has a gate to prevent unauthorized entry to the area and public contact. The disposal field is not close to any property boundary. The edge of the sprayfield and leachfield has a setback distance of 2000 feet from the ephemeral stream drainage, lakes, and flowing streams in the area.

The Spryfield/leachfield has no domestic groundwater wells. The leachfield system is designed with various effluent laterals and controls by an alternating discharge of effluent flows to the disposal areas. There are five leachfield zones. Each leachfield zone has one observation port located at the end.

There is adequate room adjacent to the existing sprayfield and leachfield for 100 percent replacement, still meeting the required setbacks. In general, storm water generated from rain and snowmelt events in the Red Fir area flows over land. However, the disinfected effluent discharge from the treatment plant to the sprayfield is rotated so that there is no direct effluent runoff to surface waterways. Bacteriological quality of ground water is monitored semi-annually.

Over ninety percent (90%) of the area's precipitation falls as snow, from October to April. A10-year average snow measurements at Giant Forest is 206 inches per year. Over 50 percent of the annual runoff occurs between May and June.

2.3. Recycled Water Projects

There are no recycled water projects at the Clover Creek area.

2.4. Operation and Maintenance

NPS Utilities Maintenance staff provides the operations for the day-to-day activities of the treatment plant including sewage pumping, routine, and preventive maintenance to protect the health of our residents, park users, and the environment in the Clover Creek area.

Most of the system operators meet the wastewater certification requirements to operate the WWTF under the supervision of a Chief Plant Operator (Grade 3) that observes and manages the development of operators toward their Grade 3 certifications. NPS staff is able to monitor operations, initiates repairs, manage wastewater treatments, and respond in the event of sewage spills.

3. GROUNDWATER QUALITY

3.1. Potential Threat to Groundwater Quality

The disposal area has eight 4-inch diameter PVC observation wells with a depth of 20 feet and the pipes extend 3 feet above ground with 1.5 feet thick concrete slab around the pipe. The observation wells are mostly dry to collect samples for nitrogen and total coliform organisms.

4. CALIFORNIA ENVIRONMENTAL QUALITY ACT

4.1. System Categorical Exemption Evaluation

There are no projects that require submission of this report at this time. Federal Government lands conduct the National Environmental Policy Act (NEPA) and the National Historic Preservation Act (NHPA) Section 106 analysis and complete the California Environmental Quality Act (CEQA) crosswalk when applicable.

5. ADDITIONAL TECHNICAL REPORTS

5.1. Sludge Management Plan

The Clover Creek WWTF generated about 19 tons of sludge in 2018/2019. A contract hauler, Western Solid Waste (Ponderosa), transports the sludge containers from the park to the Central Valley Composting facility in Dos Palos, California.

ATTACHMENTS

Attachment A – Location and Facility Map

Attachment B – Collection System Layout

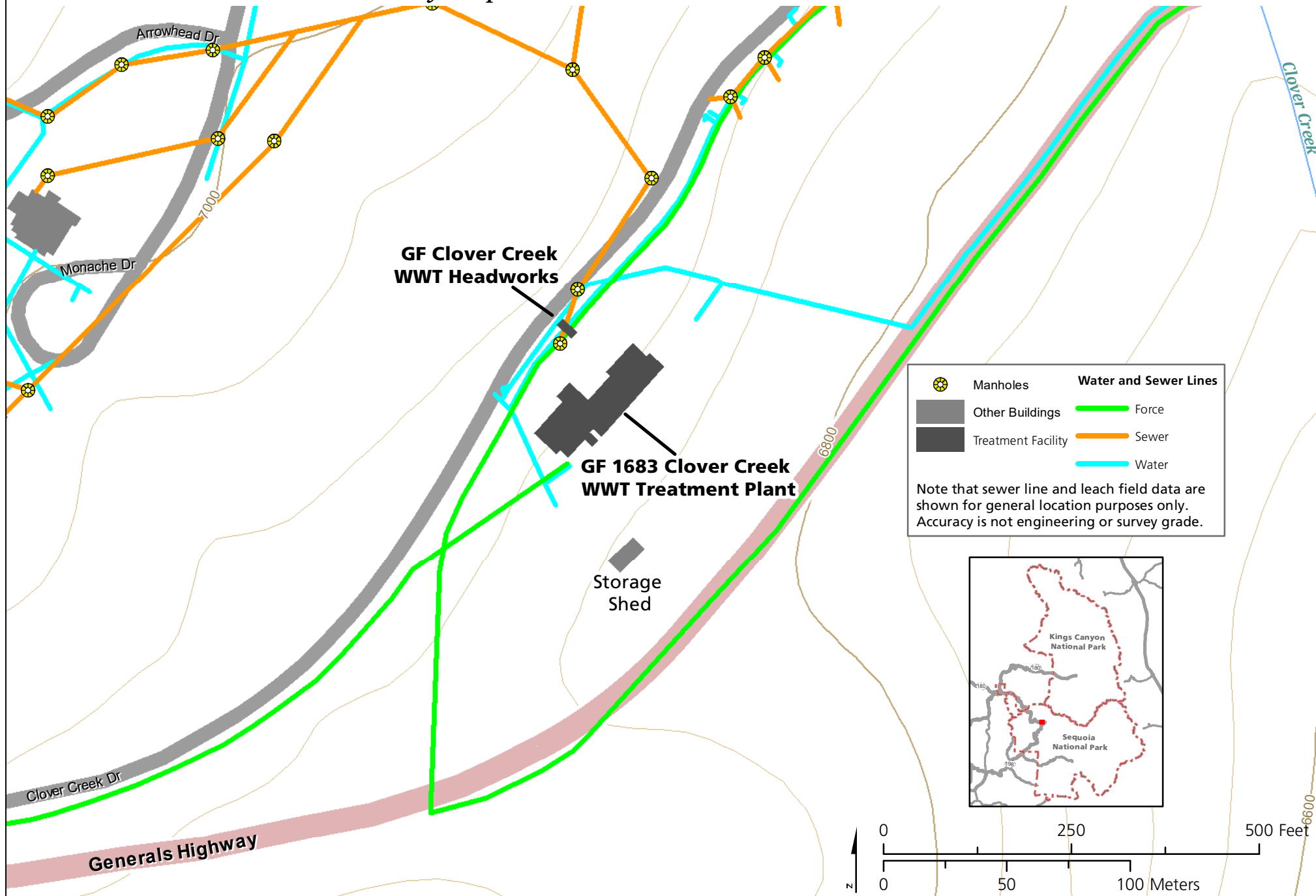
Attachment C – Wastewater Treatment Flow Schematic

Attachment D – Nitrogen Effluent Limit Evaluation



Attachment A: Location and Facility Map

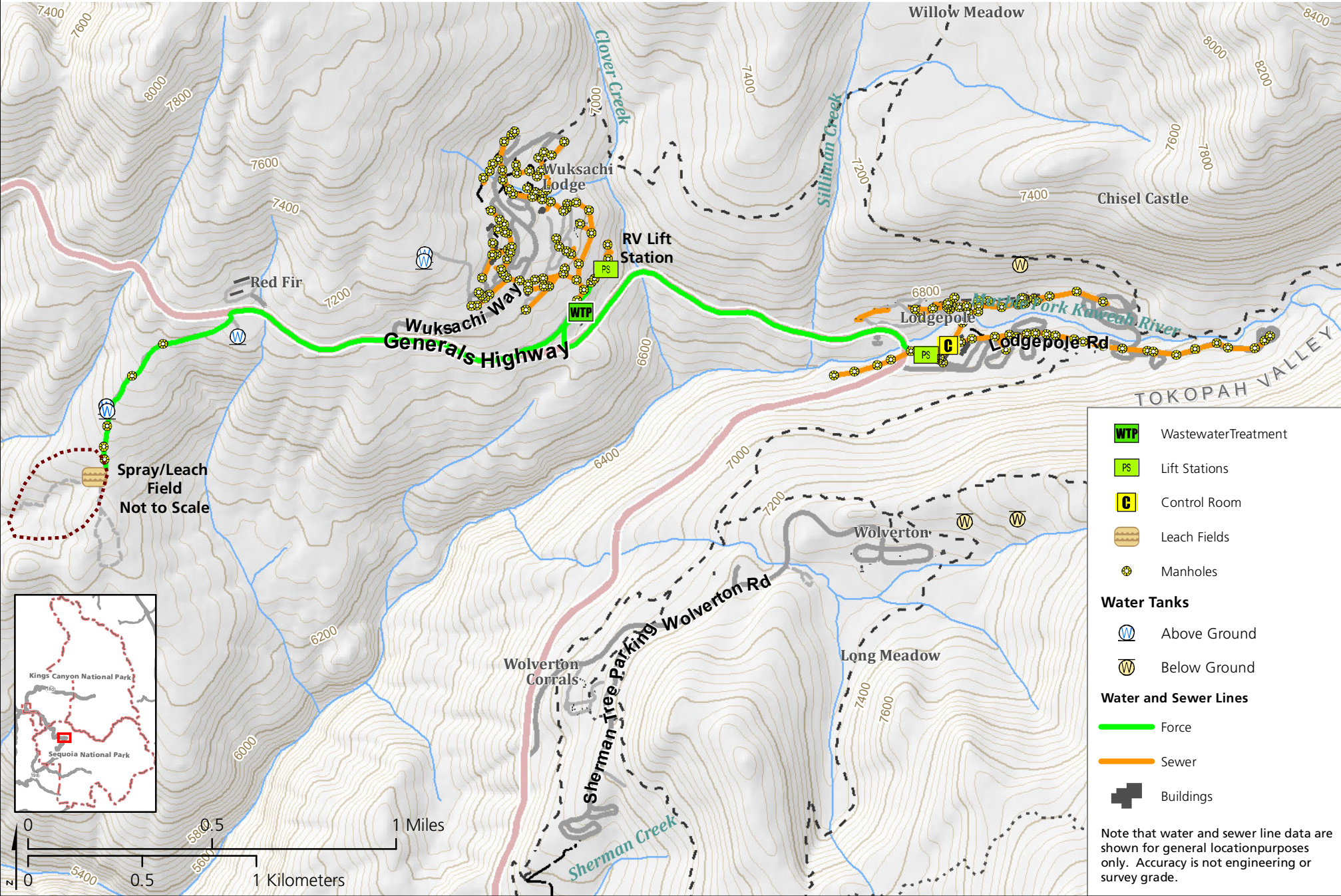
Clover Creek Wastewater Treatment Facility



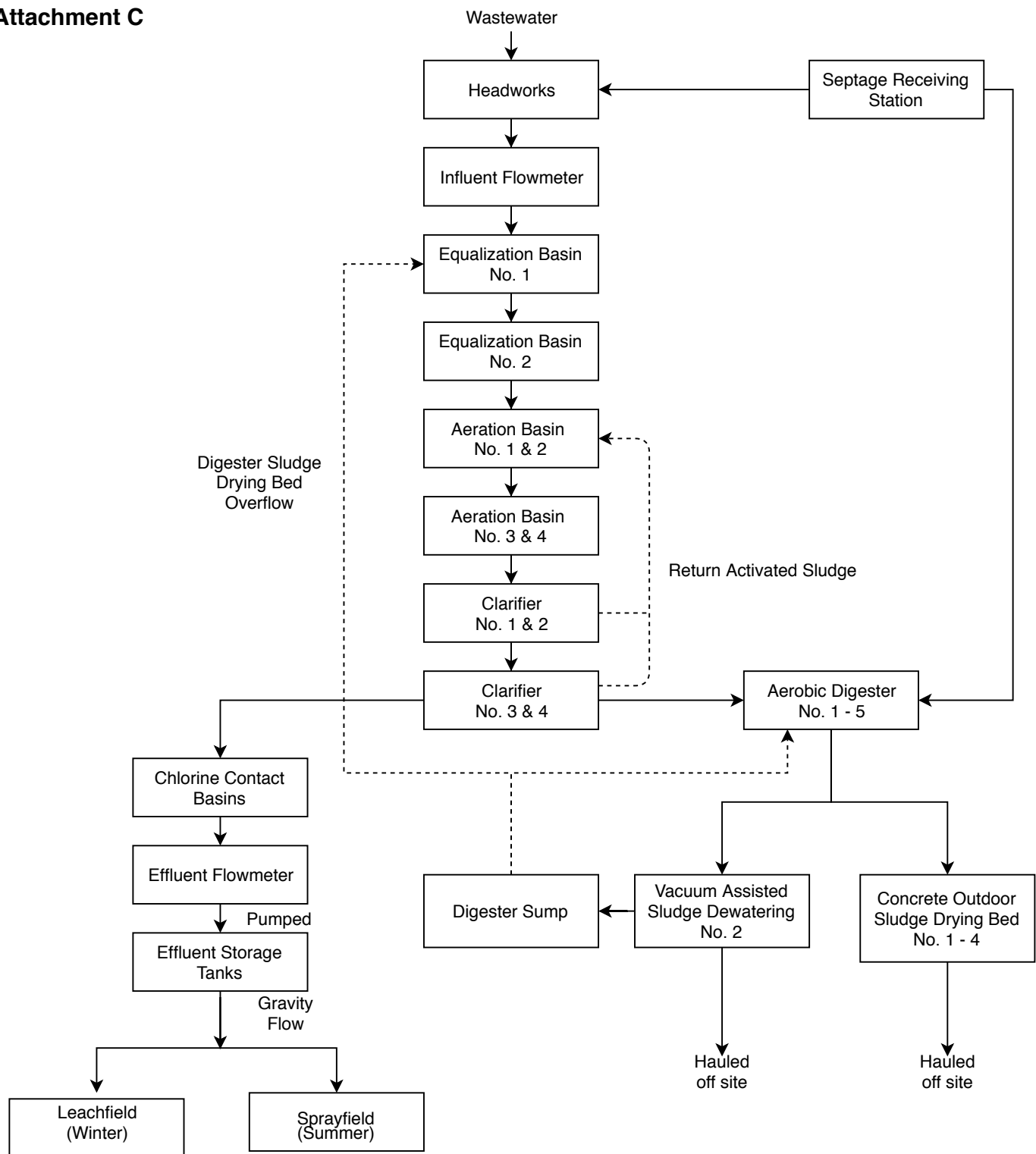


Attachment B: Collection System Layout

Clover Creek Wastewater System



Attachment C



NOT TO SCALE

Wastewater Treatment Flow Schematic
 Sequoia and Kings Canyon National Parks
 Clover Creek Wastewater Treatment Facility

Attachment D

NITROGEN EFFLUENT LIMIT EVALUATION

1. INTRODUCTION

Attachment 1 of the General Order requires five site-specific considerations to evaluate the need for nitrogen. The five site-specific considerations are flow, groundwater depth, percolation rate, wastewater strength, and if nitrogen is of concern in the area. The Clover Creek WWTF is an indoor activated sludge and tertiary treatment facility. The current design flow of the Clover Creek WWTF is 95,000 gpd, exceeds the evaluation threshold of 20,000 gpd, therefore, nitrogen evaluation is required.

2. EVALUATION

Based on the 1986 percolation test results from the Clover Creek WWTF construction report, soils in the sprayfield and leachfield area were probed to depths of 45 to 50 feet.

Seasonal groundwater was found at a depth of 30 to 35 feet, and soil percolation tests performed in the disposal area have reported rates from 1 to 5 minutes per inch. The sprayfield and leachfield percolation rates and depth to groundwater evaluation meets the conditions defined in Table 5 of the General Order.

The average treated effluent analysis of biochemical oxygen demand (BOD) and total suspended solids (TSS) based on the monthly Self-Monitoring Report (SMR) for December 2018, tabulated in Table 5, range from 1.6 to 10.48 milligrams per liter (mg/L) for BOD and 5.8 to 24.75 mg/L for TSS. This meets the domestic wastewater strength conditions defined in Findings 8 of the General Order.

3. NITROGEN REDUCTION

The disposal area has eight (8) groundwater monitoring wells that are twenty (20) feet deep. The wells are above groundwater, and are dry, so the wells were not use to collect samples for nitrogen at the time of inspection.

Instead, the NPS sampled influent and effluent samples in June and July 2019 to characterize nitrogen. The total nitrogen effluent concentrations at the Clover Creek WWTF based on the sampling results, as recorded in Table 2, range from 33.9 to 74.4 mg/L. The total nitrogen reduction for the Clover Creek WWTF is up to 40 percent.

The NPS is not required to monitor, report, or manage nitrogen in the effluent of the Clover Creek WWTF under the current permit (WDR Order 88-031).

Based on the above information nitrogen monitoring may not be required under the General Order.